

Activity: Formalization in first-order logic

Model 1 Love

Let $L(x, y)$ signify “person x loves person y .” Formalize the following conditions and statements by building up from the predicate L using the quantifiers $\forall$ and $\exists$ and the boolean connectives. Then for each of the following which are sentences (no free variables) give both a *satisfying* and *falsifying* model.

1. Person x loves everybody.
2. Someone loves person y .
3. Everyone loves person y .
4. Everyone loves somebody (but not necessarily the same single person).
5. Someone loves everybody.
6. There is somebody who is universally loved.
7. Everyone is loved by someone (but not necessarily the same single person).
8. Everybody loves themselves.
9. Love is reciprocal. (If any person loves another, that person loves them back.)
10. Everyone loves someone who loves them back.

Model 2 Divisibility

Divisibility is a very important binary relation on natural numbers. If m and n are natural numbers we say m *divides* n if m “goes perfectly into” n , i.e., when we divide n by m , the remainder is zero. Another way to say it is that n is a multiple of m .

The symbol for divides is a vertical slash, written $|$. So “ m divides n ” is written $m | n$. Many students think that $m | n$ means “ n divides m .” I don’t know why this is, but please make a note of this common error. So $2 | 6$ is **true**, because 2 divides 6 (because $2 \cdot 3 = 6$), but $6 | 2$ is **false**, because 6 does not divide 2.

11. Define $m | n$ using a logical formula which only uses multiplication $\cdot$, equality $=$, and the existential quantifier $\exists$.
12. Define “ n is even” using only 2 and $|$.

13. Define “ n is odd” using only 1, 2, $|$, and $+$.
14. Which numbers n satisfy $n \mid 0$? Which numbers n satisfy $0 \mid n$?
15. Which numbers n satisfy $n \mid 1$? Which numbers n satisfy $1 \mid n$?
16. Define $C(x, y, z)$, “ x is a common divisor of y and z .”
17. Write the divisors of 24 and 56. Which ones are common, and which is the greatest common divisor?
18. Go through the same exercise with 30 and 48.
19. What relationship do you notice that the greatest common divisor has amongst all common divisors, besides the fact that it is larger than all the rest?
20. Use your observations to write a formula defining “ x is the *greatest* common divisor of y and z .” Only use $\forall$, $|$, $\wedge$ (AND), as well as the already-defined ternary condition C . (Ternary means “has three arguments.”)

Model 3 Eventualities

Computer scientists care greatly about eventualities. For example, we might want to know that, “for all inputs x , program p *eventually* halts.” Or we might want to know, “for all sufficiently large natural numbers n , this sorting algorithm takes at most n^2 steps on lists of length n . (Talking about sufficiently large quantities is a way of expressing eventuality for “non-time-like” inputs.)

In other words, for some condition to eventually hold, it doesn’t have to hold *now*, but it has to hold *at some point*, and forevermore thereafter. Knowing that something eventually holds is often more important than knowing exactly when that occurs. By combining the power of the universal and existential quantifiers, we can formalize eventuality.

21. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function from the real numbers to the real numbers. Define the condition $P(f) \equiv$ “ f is always positive.”
22. Define “ f is positive for all inputs greater than -100 .”
23. Define “ f is positive for all inputs greater than 0.”
24. Define “ f is positive for all inputs greater than 100.”
25. Define “ f is eventually positive,” i.e., “there is *some* point past which f is always positive.”
26. Define “ f is eventually greater than 1.”

27. Define “ f is eventually greater than 1000.”

28. Define “ f eventually is greater than every number.” In other words, to ensure that f lies above any horizontal line, we just have to go sufficiently far to the right. For those of you who know calculus, this is the same as saying $\lim_{x \rightarrow \infty} f(x) = \infty$.

29. The reason there are fancy words like “monotone” instead of “increasing” is that there are multiple ways of making the notion of “ f is an increasing function” formal! One is indeed “ f is monotone,” which is defined by

$$(\forall x, y)(x < y \rightarrow f(x) < f(y)).$$

Another is $\lim_{x \rightarrow \infty} f(x) = \infty$, which we defined in the previous problem. Are these two conditions equivalent? Does either entail the other?

30. Let $M(f)$ be the condition “ f is monotone,” $A(f)$ be “ f is antitone,” and $L(f)$ be

$$\lim_{x \rightarrow \infty} f(x) = \infty.$$

Complete the analogy:

$$M : A :: L : ?$$

and give a formal definition of the missing condition.